

IMPACT OF PRICING STRATEGIES ON SALES PERFORMANCE AND CONSUMER BUYING BEHAVIOR

Project Objective

The objective of this project is to analyze and evaluate how different pricing strategies influence sales performance and shape consumer buying behavior. Specifically, the study aims to:

- Examine the relationship between pricing models (e.g., penetration pricing, premium pricing, discount strategies) and overall sales growth.
- Assess how consumer perceptions of price affect purchase decisions, brand loyalty, and value perception.
- Identify which pricing approaches maximize profitability while maintaining customer satisfaction.
- Provide actionable insights for businesses to design effective pricing strategies that balance competitiveness, revenue generation, and consumer trust.

Dataset Overview

- Number of rows → 22,000
- Number of columns
- Time range (Order_Date)
- Business context (E-commerce pricing strategy)

Primary Columns You Will Use:

- Pricing → Original_Price, Discount_%, Final_Price
- Sales → Quantity, Revenue
- Profit → Cost, Profit
- Customer → Age, Gender
- Business → Region, Category, Season

Insights:

- Overall sales performance
- Distribution of pricing strategies

Data Profiling

What you do:

- Check:
 - Mean, median, min, max
 - Unique values
 - Null values
 - Distribution

Tools:

- Python (Pandas .describe())
- SQL (GROUP BY, COUNT)
- Excel (Pivot tables)

Insights:

- Which category sells most
- Which region performs best
- Price distribution

• Data Types

Column Name	Data Type		Description
Order_ID	Integer		Unique order number
Customer_ID	Integer		Unique customer ID
Product_ID	Integer		Unique product ID
Category	String (Categorical)		Product category
Original_Price	Float		Base price before discount
Discount_%	Integer / Float		Discount percentage applied
Final_Price	Float		Price after discount

Quantity	Integer		Number of items purchased
Order_Date	Date / Datetime		Transaction date
Revenue	Float		Total sales value
Cost	Float		Cost of product
Profit	Float		Revenue - Cost
Customer_Age	Integer		Age of customer
Customer_Gender	String (Categorical)		Male / Female
Region	String (Categorical)		North, South, etc.
Payment_Method	String (Categorical)		Mode of payment
Month	Integer		Extracted from date
Season	String (Categorical)		Festive / Regular
Adjusted_Revenue	Float		Revenue with seasonal effect

KPI Tree



Data Quality Audit Report

Completeness

- Missing values in:
 - Discount (%)
 - Competitor Price
 - Customer Segment

Accuracy

- Incorrect values found:
 - Negative prices
 - Discount > 100%

Consistency

- Inconsistent formats:
 - Region names (e.g., “South”, “south”, “SOUTH”)
 - Pricing Strategy naming variations

Validity

- Data outside acceptable range:
 - Sales Quantity < 0
 - Price = 0 for non-free products

Uniqueness

- Duplicate Product_ID and Order entries detected

Issue Type	Description	Impact
Missing Values	Null values in key columns	Affects analysis accuracy
Outliers	Extremely high prices/quantities	Skews insights
Duplicates	Repeated records	Inflated sales
Invalid Data	Negative or zero values	Incorrect KPIs
Inconsistent Data	Different naming formats	Grouping errors

Root Cause Analysis

Missing Values

Duplicate Records

Invalid Data (negative price, >100% discount)

Inconsistent Formats

Outliers

Pricing Strategy Recommendations

Optimize Discount Levels

Recommendation:

- Maintain moderate discounts (10%–30%)
- Avoid deep discounts (>50%)

Insight from dataset:

- High discounts → increase sales volume
- But → reduce profitability

Adopt Dynamic Pricing Strategy

Recommendation:

- Adjust prices based on:
 - Demand trends
 - Competitor pricing
 - Seasonal variations

Insight:

- Static pricing misses revenue opportunities

Competitor-Based Pricing

Recommendation:

- Use Competitor_Price column actively
- Price slightly below or match competitors

Insight:

- Customers are price-sensitive (especially in India)

Segment-Based Pricing

Recommendation:

- Customize pricing for:
 - Customer Segment (Premium / Budget)
 - Region

Insight:

- Different customers respond differently to pricing

Focus on High-Performing Categories

Recommendation:

- Increase inventory & marketing for:
 - High-demand categories
- Reduce focus on low-performing products

Region-Based Strategy

Recommendation:

- Identify top-performing regions
- Apply region-specific pricing

Insight:

- Demand varies by location